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Needle assembly for liquid applicator

Abstract

A needle assembly for a liquid applicator comprises a housing, a needle bundle, and an elastic tubular member. The housing comprises a longitudinal channel having upper and lower open ends. The needle bundle is movably mounted in the channel and configured to be driven by a driving shaft to reciprocatively move between a retracted position and an extended position. The tubular member comprises a first end attached to the housing and a second end attached the needle bundle to bias the needle bundle to move longitudinally towards the retracted position. The tubular member comprises a thicker first portion and a thinner second portion separated by a slanted boundary, which extends between the first and second ends in both the longitudinal direction and a lateral perimetral direction. The slanted boundary may be helical. The first portion may include a pair of symmetrically arranged slanted ridges.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2015/0151098	12/2014	Spendlove et al.	N/A	N/A
2020/0114137	12/2019	Siciliano	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101587235	12/2015	KR	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part application of, and claims the benefit of, and priority to, U.S. patent application Ser. No. 17/682,831, filed 28 Feb. 2022, entitled “Needle Assembly for Liquid Applicator”, the entire contents of which are incorporated herein by reference. (2) This application also claims the benefit of, and priority to, European Patent Application No. EP23158470.7, filed 24 Feb. 2023, entitled “Needle Assembly for Liquid Applicator,” which in turn claims priority from U.S. patent application Ser. No. 17/682,831. The entire contents of EP23158470.7 are incorporated herein by reference. (3) This application further claims the benefit of, and priority to, Chinese Patent Application No. 202310143462.8, filed 10 Feb. 2023, entitled “Needle Assembly for Liquid Applicator,” which in turn claims priority from U.S. patent application Ser. No. 17/682,831. The entire contents of CN 202310143462.8 are incorporated herein by reference.

FIELD

(1) The present invention relates generally to devices for applying a liquid to skin, particularly to

needle assembly used in liquid applicators such as tattoo devices or devices for applying permanent make-up.

BACKGROUND

(2) Liquid applicators for applying ink to skin, such as tattoo devices, typically include a needle for applying ink to skin, a base with a needle actuator, and a needle handle that connects the needle to the base and can be conveniently held in a hand of an operator for manipulating the needle during use. In operation, the needle is actuated by the needle actuator to reciprocally move between extended and retracted positions, thereby repeatedly puncturing the subject's skin. Tattoo needles may be provided in a needle assembly, which is often referred to as needle module and is detachably attached to the needle handle. Examples of tattoo devices include rotary tattoo machines and coil tattoo machines. Some needle modules include an elastic biasing member for sealing and biasing the needles both axially and transversally.

(3) For example, U.S. Pat. No. 11,052,232 to Xiao, issued Jul. 6, 2021, disclosed a needle assembly with an elastic biasing member for biasing needles and forming a fluid seal.

(4) CN106902452A by Wang, published Jun. 30, 2017, disclosed a tattoo device, which includes a shell, a needle, and an elastic membrane. A linkage structure is arranged on the needle and fixedly connects the needle with the elastic membrane. The elastic membrane has regions of different thicknesses for generate forces to move the needle backwards and towards an operation section. The elastic membrane may have an elastic rib extending along the direction of the needle movement on one side of the elastic membrane.

SUMMARY

(5) The present disclosure relates to needle assemblies for liquid applicators. The needle assembly comprises a housing, a needle bundle, and an elastic tubular member. The housing comprises a longitudinal channel having upper and lower open ends. The needle bundle is movably mounted in the channel and configured and mounted to be driven by a driving shaft to reciprocally move between a retracted position and an extended position. The tubular member comprises a first end attached to the housing and a second end attached the needle bundle to bias the needle bundle to move longitudinally towards the retracted position.

(6) In a first aspect, the elastic tubular member comprises a helical ridge extending on an external surface of the tubular member such that the helical ridge provides a biasing force that biases the needle bundle both longitudinally towards the retracted position and laterally towards a lateral side of the channel of the housing.

(7) In different embodiments, the needle assembly may include one or more of the following features. The first and second ends of the tubular member may be sealingly attached to the housing and the needle bundle to prevent fluid communication between the lower open end and the upper open end of the housing. The tubular member may comprise a diaphragm. The ridge may extend from the first end to the second end of the tubular member. The helical ridge may comprise two helical ridges arranged symmetrically with respect to a plane passing through a longitudinal axis of the tubular member. The helical ridge may comprise helically aligned step-shaped side edges. The helical ridge may comprise generally straight side edges. The helical ridge may comprise curved side edges. The helical ridge may comprise two generally parallel side edges or may comprise unparallel side edges. The helical ridge may comprise a plurality of helically arranged separate ridge sections. The helical ridge may have a generally polygonal or semicircular cross-section. The polygonal cross-section may be triangular, rectangular, or trapezoidal. The tubular member may comprise silicone, latex, or rubber.

(8) In another aspect, the tubular member comprises a first portion and a second portion thinner than the first portion, which portions are separated by a boundary therebetween. The boundary between the first and second portions extends helically between the first and second ends of the tubular member. As a result, the tubular member when stretched exerts a biasing force on the needle bundle that biases the needle bundle both longitudinally towards the retracted position and

laterally towards a lateral side of the channel in the housing.

(9) In different embodiments, the needle assembly may include one or more of the following features. The first and second ends of the tubular member may be sealingly attached to the housing and the needle bundle to prevent fluid communication between the lower open end and the upper open end of the housing. The first and second portions of the tubular member may be separated by two helical boundaries, arranged symmetrically with respect to a plane passing through a longitudinal axis of the tubular member. The second portion of the tubular member may comprise a bellows section. The first and second portions of the tubular member may be separated by a helically aligned step-shaped side wall or a curved side wall at the boundary. The tubular member may comprise silicone, latex, or rubber.

(10) In a further aspect, the tubular member comprises a first portion and a second portion thinner than the first portion. A slanted boundary between the first portion and the second portion extends between the first end and the second end in both the longitudinal direction and a lateral perimetral direction to provide a lateral biasing force that biases the needle bundle laterally toward a lateral side of the channel. In different embodiments, the slanted boundary may comprise a helical or straight boundary. The first portion may comprise a slanted ridge extending on an external surface of the tubular member such that the slanted ridge provides a biasing force that biases the needle bundle laterally toward a lateral side of the channel. The slanted edge may be a helical ridge. The slanted ridge may extend from the first end to the second end of the tubular member. The slanted ridge may comprise two ridges arranged symmetrically with respect to an axial plane passing through a longitudinal axis of the tubular member. The slanted ridge may have a generally polygonal or semicircular cross-section. The first and second portions of the tubular member may be separated by a pair of slanted boundaries, arranged symmetrically with respect to an axial plane passing through a longitudinal axis of the tubular member. The second portion of the tubular member may comprise a bellows section. The first portion and the second portion of the tubular member may be separated by a step-shaped side wall or a curved side wall. The first and second ends of the tubular member may be sealingly attached to the housing and the needle bundle to prevent fluid communication between the lower open end and the upper open end of the housing.

(11) In different embodiments, the elastic tubular member may have a generally cylindrical, frustum, or prismatic profile. An external lateral perimeter of the tubular member may be generally polygonal or elliptical. The tubular member may comprise silicone, latex, or rubber.

(12) As can be appreciated, the slanted boundary or ridge on an external surface of the elastic tubular member provides a biasing force that has both a longitudinal component and a lateral component.

(13) Other aspects, features, and embodiments of the present disclosure will become apparent to those of ordinary skill in the art upon review of the following description of specific embodiments in conjunction with the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the figures, which illustrate, by way of example only, embodiments of the present disclosure:

(2) FIG. 1 is a cross-sectional view of a needle assembly, illustrative of an embodiment of the present disclosure;

(3) FIG. 2 is a perspective view of the needle assembly of FIG. 1;

(4) FIGS. 3 and 4 are exploded perspective views of the needle assembly of FIG. 2;

(5) FIGS. 5A and 5B are partially cross-sectional and partially elevation views of the needle assembly of FIG. 1, with the needle bundle in the retracted position (5A) or the extended position (5B) respectively;

- (6) FIG. 6 is a force diagram showing the biasing force and its longitudinal and lateral component applied by the tubular member in the needle assembly of FIG. 5B;
- (7) FIG. 7A is a front elevation view of a tubular member, according to an embodiment of the present disclosure;
- (8) FIG. 7B is a bottom view of the tubular member of FIG. 7A, showing the symmetry plane indicated by line B-B;
- (9) FIG. 7C is a top perspective view of the tubular member of FIG. 7A;
- (10) FIG. 7D is a cross-sectional view of the tubular member of FIG. 7A along line 7D-7D in FIG. 7A;
- (11) FIG. 8A is a bottom perspective view of the tubular member of FIG. 7A;
- (12) FIG. 8B is a perspective view of a cut-off portion of the tubular member of FIG. 7A, which is cut along line 8B-8B in FIG. 8A;
- (13) FIGS. 9A-9G are elevation views of different tubular members, according to different embodiments of the present disclosure;
- (14) FIGS. 9H and 9I are elevation views with partial sectional views of different tubular members, according to further different embodiments of the present disclosure;
- (15) FIG. 10A is a perspective view of a tubular member, illustrative of another embodiment of the present disclosure;
- (16) FIG. 10B is a perspective view of a cut-off portion of the tubular member of FIG. 10A, taken along line 10B-10B in FIG. 10A;
- (17) FIG. 11A is a cross-sectional view of the helical ridge of the tubular member of FIG. 7A, taken along line 11A-11A in FIG. 7A; and
- (18) FIGS. 11B to 11F are cross-sectional views of different helical ridges, according to different embodiments of the present disclosure.

DETAILED DESCRIPTION

- (19) An embodiment of the present disclosure relates to needle assemblies for use in liquid applicators for applying a liquid to skin.
- (20) The liquid applicator may be an ink applicator, such as tattoo devices as described in U.S. Pat. Nos. 11,052,232; 10,806,915; or US 20190217072, the entire contents of each of which are incorporated herein by reference. Briefly, a suitable tattoo device (not shown) may include a needle module coupled by a needle handle (not shown) to a base device (not shown). The needle module may be a disposable needle module. The needle module and the needle handle may be configured to be removably coupled to each other.
- (21) In an embodiment disclosed herein, the needle assembly may be a needle module **10** as illustrated in FIGS. **1** to **8B**. The needle module **10** may be configured to apply tattoo or permanent make-up to skin. The needle module **10** may be a disposable module.
- (22) As illustrated in FIG. **1**, needle module **10** includes housing **40** and a reciprocally movable needle bundle **200** with a needle tip portion **210** mounted in the housing **40** for applying ink to the skin of a subject.
- (23) While not shown in the drawings, it should be understood that the needle module **10** can be attached or connected to a tattoo machine (not shown), such as through a handle (not shown) of the tattoo machine. A base device (not shown) of the tattoo device includes a needle actuator (not shown) with a drive shaft (not shown) for actuating downward movement of the needle bundle **200** through a needle shaft portion **220** of the needle module **10**.
- (24) The construction and operation of a suitable tattoo machine, including its handle and base device, are known in the art, and can be similar to, for example, the corresponding devices and parts disclosed in U.S. Pat. Nos. 11,052,232, 10,806,915, or US 20190217072. As such, these devices, parts, and their operation, and their coupling and interaction with housing **40** and needle shaft portion **220** are not described in detail herein.
- (25) As depicted in FIGS. **1-5B**, the needle bundle **200** is movably mounted in needle housing **40**.

The needle housing **40** includes a tubular body portion **100** and a removable cap **400**.

(26) The body portion **100** has a tubular longitudinal inner channel **120** for receiving and housing the needle bundle **200**. The channel **120** extends from an upper open end **122** to a lower open end **124** of the body portion **100**, and has a central longitudinal axis (A), as shown in FIG. 4 (also see FIG. 5A). The body portion **100** has an opening **150** at the lower open end **124** to allow the needle tip portion **210** of the needle bundle **200** to pass through. A needle guiding wall **110** is provided at the lower open end **124** adjacent the opening **150**, which provides a guide surface for guiding movement of the needle tip portion **210**.

(27) The opening **150** may be provided as an integral part of the body portion **100** as depicted in the drawings. However, in other embodiments, the opening **150** may be provided in a separate mouthpiece (not shown) attached to the body portion **100**. The mouthpiece can be constructed and provided according to known techniques, including those disclosed in the references incorporated herein.

(28) The body portion **100** may also have one or more internal coupling ribs **130** at the upper open end **122** for coupling with and engaging the cap **400**, as will be further described below.

(29) The removable cap **400** is provided at and coupled to the upper open end **122** of the tubular body portion **100**. The cap **400** has a central opening **450** for allowing the needle shaft portion **220** to pass therethrough and axially move up and down during operation. Cap **400** also has a downward extending tube **410**, with an external annular rim **420** and a lower external annular groove **430**, which are provided and configured for engaging an upper end portion **320** of the tubular member **300** to affix the upper end portion **320** to the housing **40**, as will be further described below. The cap **400** also has an upper annular groove **440** for engaging coupling ribs **130** of the body portion **100**, so as to secure the cap **400** in position.

(30) Each or both of body portion **100** and cap **400** may be made of a plastic material, or any other suitable material.

(31) As already alluded to above and illustrated in FIGS. 1-5, the needle bundle **200** has a needle tip portion **210** attached to a needle shaft portion **220**.

(32) In various embodiments, the needle tip portion **210** of the needle bundle **200** may include one or more needle tips, which may be welded together, or otherwise bounded together. The needle tips may be formed of stainless steel or any other suitable material. The individual needle tip may have any suitable or known needle tip shape. The needle tips may be arranged, such as being tied together in an array, to form a tip portion that has a generally or substantially cylindrical or conical profile. Alternatively, an array of the needle tips may be arranged side-by-side to form a tip portion that has a generally flattened or band-shaped profile. Rows of side-by-side needles may also be stacked. Such different arrangements of tattoo needles are known in the art and may be referred to as “Round Liner” needles, “Round Shader” needles, “Flat” Needles, or “Magnum” needles, respectively. Needle tip portion **210** may include 1-18 individual needle tips for “Round Liner” needles and “Round Shader” needles, or may include 4-27 needle tips for “Flat” needles and “Magnum” needles.

(33) As can be appreciated, the cross-sectional sizes or diameters of the needles or needle bundles will affect how the ink will flow. Typically, the smaller the needle tip size or narrower the diameter of the needle tip, the finer and more controlled the stream of ink that flows off each needle tip. Typically, the needle tips in the same needle bundle may be of the same or similar sizes. The size of the needle tips may be selected based on the desired effects by the operator or user. Different sizes may be used for different reasons. Standard sizes of needles may be used. The diameters of the individual needles may be 0.25 mm, 0.30 mm, or 0.35 mm in some embodiments. The designs of the needle tips may be selected and vary as known in the art based on the desired tattooing techniques and purposes to be applied.

(34) The number of needles in a needle bundle **200** may vary from 1 to 27 or more as desired. For example, commercially available round needle bundles typically have 1, 3, 5, 7, 8, 9, 11, 14, or 18

needles in each bundle. It can also be appreciated that the overall profile of the needle bundle may change and vary depending on the number of needles in the bundle, their arrangement, the amount of soldering material used, or other factors.

(35) The needle tips in needle tip portion **210** are supported on needle shaft portion **220**. The axis of needle shaft portion **220** may be axially aligned with the axis of channel **120** (axis A shown in FIG. **4** and FIG. **5A**), but is off-set from the axis of the needle tip portion **210** (see FIGS. **3**, **4** and **5A**). The needle shaft portion **220** may be formed of a rigid plastic material or another suitable material. The material in needle shaft portion **220** should have relatively high strength and high hardness and be wear-resistant so that its sufficiently rigid and strong to support stable movement of the needle tip portion **210** during operation.

(36) In the embodiment as depicted in FIG. **1**, the needle tip portion **210** is attached to needle shaft portion **220** through an optional connection rod **215** of the needle bundle **200**. Connection rod **215** may be made of steel such as stainless steel. An end of the connection rod **215** may be attached by an adhesive to needle shaft portion **220** in a bore **222** in the needle shaft portion **220**. The needle tip portion **210** may be attached to the other end of the connection rod **215** such as by soldering. The connection rod **215** may have a diameter of about 1.5 mm, and the bore **222** in the needle shaft portion **220** may have a corresponding diameter.

(37) In an alternative embodiment, the needle tip portion **210** may have a cylindrical end which is directly inserted into the bore **222** of the needle shaft portion **220** for attachment to the needle shaft portion **220** such as by an adhesive. However, when the needle tip portion **210** is attached to the needle shaft portion **220** through the connection rod **215**, the needle tip portion **210** may have different sizes or different number of needles, and it is not necessary for the needle tip portion **210** to have a cylindrical end with the shape and size that match the shape and size of the bore **222**.

(38) Needle bundle **200** may reciprocally move longitudinally along axis A between the retracted position, illustrated in FIG. **5A**, and the extended position, illustrated in FIG. **5B**. As noted, the downward movement of needle bundle **200** is actuated by the actuator of the tattoo machine to which the needle bundle **200** is connected.

(39) To facilitate the reciprocal movement of the needle bundle **200**, an elastic tubular member **300** is provided within housing **40** and configured to pull the needle bundle **200** upward during each stroke cycle after the needle bundle **200** has been pushed down by an actuating or driving mechanism such as the drive shaft of the tattoo device described above.

(40) The tubular member **300** is also configured to bias the needle tip portion **210** of the needle bundle **200** laterally towards a lateral side of the channel **120**, such as the needle guiding wall **110**.

(41) For ease of reference, it is assumed that the tubular member **300** has a central or longitudinal axis extending along the longitudinal direction, a sagittal plane, a frontal plane, and a transverse plane. As shown in FIG. **1**, the axial or longitudinal direction is up and down, the sagittal plane is the plane in which the cross-sectional view of FIG. **1** is taken; the frontal plane is perpendicular to sagittal plane and along the axis with the direction facing the left hand side being the front direction (and the opposite direction being the rear direction); and the transverse plane is perpendicular to both the sagittal plane and the frontal plane. The central axis is also shown along axis A in FIGS. **4** and **5A**. The sagittal plane is along the line B-B shown in FIG. **7B**. The frontal plane is along the line 7D-7D shown in FIGS. **7A** and **7B**.

(42) Furthermore, tubular member **300** is configured and assembled to provide a fluid seal between the lower open end **124** and the opening **450** of the cap **400** in the housing **40**. The tubular member **300** may thus form a fluid seal between the lower open end **124** and the opening **450**.

(43) Conveniently, when a fluid, such as ink or blood or other bodily fluid, enters housing **40** through opening **150**, the seal provided by tubular member **300** will prevent the fluid from passing through opening **450** and from contaminating the upper components in the housing **40** including the upper section of the needle shaft portion **220** that will extend to outside the opening **450** during operation. Consequently, the fluid is contained within housing **40** and contamination of components

of the tattoo device connected to the needle module **10** can be prevented. The fluid in the housing **40** may be disposed with the needle module **10** when the needle module **10** is discarded after use. For a new subject to be tattooed, a new needle module **10** may be used and connected to the tattoo device to prevent cross-contamination through the needle module.

(44) As can be better seen in FIGS. 7B-7D and 8A, the tubular member **300** includes a lower end portion **310**, an upper end portion **320**, and a middle, tubular section **330** connecting the upper end portion **320** and lower end portion **310**.

(45) The upper end portion **320** of tubular member **300** is sized and shaped to close fit with the inner wall of tubular body portion **100**. The upper end portion **320** has annular ridges extending along the circumference of the outer wall of upper end portion **320**, which provide fluid sealing and frictional fitting with the inner wall of tubular body portion **100**.

(46) As noted earlier, cap **400** is configured to engage the upper end portion **320** of tubular member **300**. In particular, the upper end portion **320** is securely retained between cap **400** and tubular body portion **100** in annular groove **430** by annular rim **420**, and is thus sealingly affixed to housing **40**.

(47) The lower end portion **310** of tubular member **300** is configured to sealingly engage the needle shaft portion **220** of needle bundle **200**. As depicted in FIG. 8A, the lower end portion **310** of tubular member **300** may include a central opening **312** for allowing the needle shaft portion **220** to pass through and frictionally engaging the needle shaft portion **220** with fluid sealing.

(48) The middle tubular section **330** has a tubular shape and includes a helical ridge **340**, or wall sections of different thickness separated by a helical boundary, as will be further described below.

(49) Notably, the tubular member **300** in embodiments described herein and shown in the drawings is constructed differently from the biasing member disclosed in U.S. Pat. No. 11,052,232 or the biasing member disclosed in US 20190217072.

(50) Specifically, as depicted in FIGS. 3-5 and in FIGS. 5A-8B, the tubular member **300** in the illustrated embodiment includes two helical ridges **340** extending on its external surface. The two helical ridges **340** are symmetrically positioned with respect to the sagittal plane, which passes through the central longitudinal axis of the tubular member **300**, which is aligned with axis A as shown in FIGS. 4 and 5A. As shown in FIG. 7B, the symmetry plane is the sagittal plane indicated by line B-B, and will be referred to as symmetry plane B hereinafter. Conveniently, the helical ridges **340** provide biasing forces that bias the needle bundle **200** laterally toward a lateral side of the channel **120** in addition to biasing the needle bundle **200** longitudinally upwards.

(51) As used herein, a “helical ridge” refers to a ridge that extends along a cylindrical or conical external surface in both the axial or longitudinal direction and the circumferential direction. The three-dimensional (3D) profile of the ridge may form a generally geometrical helix or spiral. However, in different embodiments, a helical ridge may have other curved or straight-edged shapes that do not form a perfect helix or spiral. For example, a helical ridge may have a shape similar to that of the slanted tooth trace on a helical gear. A “helical boundary” may also refer to a boundary line that extends along a cylindrical or conical external surface in both the axial or longitudinal direction and the circumferential direction. As used herein, a “helical” line refers to a line that extends in three dimensions and is not parallel to any straight line nor within any single plane.

(52) However, in some embodiments, such as when the tubular member **300** has one or more flat sides (not shown), the helical ridge or boundary may be replaced with a different but still slanted ridge or boundary, where the ridge or boundary structure extends in both the longitudinal direction and a lateral perimetral direction along an external, perimetral surface of the tubular member **300**. The slanted boundary or ridge may be curved when the lateral external perimeter of the tubular member has a curved section, such as when the perimeter is generally elliptical or has a curved section. The slanted boundary or ridge may also be straight if the lateral external perimeter of the tubular member is generally polygonal or has a straight section. As used herein, a “lateral external perimeter” of the tubular member **300** refers to an external cross-sectional perimeter of the tubular member **300** in a transverse plane that is perpendicular to the axial or longitudinal direction of the

tubular member **300**. Thus, a line is considered “slanted” when the slanted line extends in both the longitudinal direction and the lateral perimetral direction. A slanted boundary or ridge therefore refers to an elongated boundary or ridge with a length that extends in both the longitudinal direction and the lateral perimetral direction, but does not include a single straight boundary or ridge elongated within an axial plane, or aligned symmetrically with respect to any axial plane. An axial plane is any plane that passes through the axis of the tubular member **300**, and may be, for instance, the sagittal plane of a cylinder, frustum, or prism.

(53) For clarity, it is noted that, for example, a straight line on the external surface of a conical tubular body that extends within the sagittal plane of the tubular body and does not deviate from the sagittal plane is not considered a “slanted” line within the meaning of “slanted” as used herein. Similarly, a straight ridge that extends along the sagittal plane and is symmetrically aligned with the sagittal plane is also not a “slanted” ridge within the meaning in this disclosure. A transversal line that is within the transversal plane and extends along the lateral perimeter is likewise not considered a “slanted” line herein. Thus, a wall or groove extending along the transversal line does not have a “slanted” boundary.

(54) For embodiments disclosed herein, a tubular body may have a cylindrical, frustum, or prismatic external profile, which may have a generally elliptical or polygonal lateral external perimeter. An elliptical perimeter may be circular or oval. A polygonal perimeter may be square or rectangular, or hexagonal. Tubular bodies may include cylindrical shaped bodies, and frustums of cones and pyramids, such as truncated cones or truncated pyramidal shapes. The outer surfaces of a polygonal tubular body may have flat surfaces. With a polygonal cross-section, a slanted boundary or slanted ridge on the tubular body may extend on a flat surface of the tubular body and is slanted with respect to the axial plane that is perpendicular to the flat surface. In this case, the slanted boundary or ridge on the flat surface may extend in a straight line but is still slanted with respect to that axial plane. A slanted boundary or ridge may thus be curved, such as being helical, or may extend along a straight line or have one or more straight sections. In any event, a slanted ridge or boundary provides a biasing force that has both a longitudinal component and a perimetral or latitudinal component, as well as an optional radial component. Providing a pair of symmetrically arranged corresponding slanted boundaries or slanted ridges however has the effect that when the tubular member **300**, particularly the middle tubular portion **330**, is stretched, the total biasing force applied by the tubular member **300** on the needle bundle **200** includes a larger radial component that biases the tip portion **210** of the needle bundle **200** towards the desired side of the channel **120**, such as the guiding wall **110**.

(55) In different embodiments, the elastic tubular member **300**, or its middle tubular portion **330**, may have a generally cylindrical profile, frustum profile, or prism profile, or may have a generally polygonal or elliptical external lateral perimeter or cross-section. An elliptical shape may be circular. In some embodiments, the polygonal shape may have 3 to 12 generally straight sides. That is, the external lateral perimeter of the tubular member **300** may have 3 to 12 generally straight sections. For example, the external lateral perimeter may have a generally triangular, quadrilateral, pentagonal, or hexagonal shape.

(56) The first and second ends of the tubular member are sealingly attached to the housing and the needle bundle, to prevent fluid communication between the lower open end and the upper open end of the housing. Thus, the tubular member **300** also functions as a seal. The tubular member **300** may comprise a diaphragm for this purpose.

(57) In the embodiment depicted in FIGS. 1-8B, the helical ridges **340** extend from the first end to the second end of the tubular member **300**. However, in different embodiments, one or more shorter helical ridges, or a plurality separate ridge sections arranged in a helical pattern, may be provided on tubular member **300** to provide the lateral biasing force.

(58) As depicted in FIGS. 5A, 5B, and 7A, the helical ridges **340** comprise generally straight side edges when viewed in a side elevation view. However, in different embodiments, the side edges of

the helical ridges **340** may have different shapes or profiles, as will be described in more detail below.

(59) The tubular member **300** may also be considered to include thicker portions and thinner portions, and the thicker and thinner portions are separated by a helical boundary therebetween that extends helically between the first end and the second end of the tubular member **300**. For example, in various embodiments of the tubular member **300** as shown in FIGS. 5A-8B and 9A to 9D, the helical ridges **340** are thicker portions and the regions between the helical ridges **340** are the thinner portions, and the boundaries are at the bottom edges of the helical ridges **340**.

(60) In some embodiments, a tubular member **300** may comprise only two portions, one thicker portion and one thinner portion as illustrated in FIG. 9E. The two portions are separated by two helical boundaries **345**, which may be arranged symmetrically with respect to a plane passing through a longitudinal axis of the tubular member **300**. A curved side wall may be formed at the helical boundary between the thinner and thicker portions.

(61) The thinner portion of the tubular member **300** may comprise one or more bellows sections. The tubular member **300** may be provided with one bellows section on one side of the tubular member **300**, such as at the wall **332** as illustrated in FIG. 9H, or with two bellows sections on opposite sides, such as at both the wall **332** and the opposite wall **334** as illustrated in FIG. 9I.

(62) Tubular member **300** may be formed of a silicone material or another resilient material such as latex, rubber, or an elastomer. The resilient material may be selected such that it is sufficiently elastic to be extended under stress to the desired extended position to expose the needle tips, but also sufficiently resilient to provide the required biasing force for returning the needle bundle **200** from the extended position back to the retracted position. In one embodiment, tubular member **300** may be formed from a silicone material with a Shore hardness from about 30 A to about 50 A.

(63) During use, the needle module **10** is attached to a handle of a tattoo device and the needle bundle **200** is driven by a needle actuator of the tattoo device to move downward so that the needle bundle **200** will move to the extended position in which the needle tips extend to outside of the lower opening **150** of the housing **40**. The tubular member **300** conveniently facilitates reciprocal movement of the needle bundle **200** by pulling the needle bundle **200** upwards during each movement cycle after the needle bundle **200** has been pushed downward by the actuator. The biasing force is increased as the elastic tubular member **300** is stretched further and further during the downward movement of the needle bundle **200**.

(64) Further, due to the slanted or helical profile of the ridge **340** (or the slanted or helical boundary), the tubular member **300** also biases the needle tip portion **210** of the needle bundle **200** towards one side of the housing channel **120**, such as towards the guiding wall **110** at the lower end near the opening **150**. This lateral biasing of the needle tip portion **210** towards or against the guiding wall **110** can stabilize the needle bundle and maintain steady contact between the needle bundle **200** and the guiding wall **110**.

(65) In addition, tubular member **300** conveniently provides a fluid seal between the lower open end **150** and the upper open end **450** of the housing **40**.

(66) The upper end portion **320** of tubular member **300** is configured to engage the inner wall of body portion **100** and cap **400**. For example, as best depicted in FIG. 7A, the upper end portion **320** of the tubular member **300** may include a number of annular ridges **322**, which are sized and shaped to close fit with the inner wall of the tubular body portion **100**. In the depicted example, the upper end portion **320** has 4 annular ridges extending along the circumference of the outer wall of the upper end portion **320**. The inner wall of upper end portion **320** defines an opening that allows the needle shaft portion **220** of the needle bundle **200** to pass through and for engaging the cap **400**.

(67) After assembly, the upper end portion **320** is securely retained between cap **400** and tubular body portion **100** and is sealingly affixed to housing **40**. As the upper end portion **320** of tubular member **300** is not attached to needle bundle **200**, it does not move when the needle bundle **200** moves up and down. Thus, during operation the upper end portion **320** of tubular member **300** is

fixed in position relative to the housing **40**.

(68) The lower end portion **310** of tubular member **300** is configured to sealingly engage the needle shaft portion **220** of needle bundle **200**. The lower end portion **310** has an opening **312**, which is sized and shaped such that the needle shaft portion **220** forms a close fit with the lower end portion **310** at the opening **312**, so as to form a fluid-tight seal between the tubular member **300** and the needle bundle **200**, while at the same time allowing the shaft portion **220** to axially move up and down during operation without breaking the seal.

(69) For example, as can be better seen in FIGS. **1** and **7D**, the lower end portion **310** has a thickened section with an upward facing surface, and the needle shaft portion **220** may have an enlarged section, shoulder **230**, that engages the upward facing surface of the lower end portion **310** of the tubular member **300**. The thickened section of the lower end portion **310** is sized and shaped to closely couple with the shoulder **230**, such that shoulder **230** exerts a downward force on the thickened section of the lower end portion **310** to pull the lower end portion **310** down when the needle bundle **200** is driven by the needle actuator to move towards the extended position. As the lower end portion **310** is pulled downward, the tubular member **300** is stretched and the length of the tubular member **300** increases. As a result, the tubular member **300** produces a biasing force to bias the needle bundle **200** upward. After the needle actuator stops driving the needle bundle **200** downward and the needle bundle **200** reaches the final extended position, the biasing force applied by the tubular member **300** pulls the needle bundle **200** back up to the retracted position.

(70) To further limit the relative movement between the lower end portion **310** of tubular member **300** and the needle bundle **200**, the needle bundle **200** may optionally include a collar **232** spaced from the shoulder **230** by a neck section **234** therebetween, so that the lower end portion **310** of the tubular member **300** can be received in the groove formed by the shoulder **230**, collar **232** and neck section **234** to attach the lower end portion **310** to the needle bundle **200**. Thus, the movement of the lower end portion **310** relative to the needle bundle **200** is limited and restricted by shoulder **230** and collar **232**.

(71) In different embodiments, the tubular member **300** may be formed of an elastic and resilient material, which may be selected such that it is sufficiently elastic to be extended under stress to the desired extended position to expose the needle tips, but also sufficiently resilient to provide the required biasing force for returning the needle bundle **200** from the extended position back to the retracted position, and for applying an adequate lateral biasing force to press the needle tip portion **210** of the needle bundle to a selected side of the channel in the housing **40**, such as the guiding wall **110** as illustrated in FIG. **3**. In some embodiments, the resilient material may be a silicone material with a Shore hardness from about 25 A to about 50 A.

(72) In the embodiment depicted in the drawings, the upper end portion **320** and lower end portion **310** each has a wall thickness that is greater than the wall thickness of the middle tubular section **330**. The upper end portion **320** and lower end portion **310** may be more rigid, less elastic, and less resilient, than the middle tubular section **330**.

(73) FIGS. **5A** and **5B** show the needle module **10** with the needle bundle **200** in the retracted (FIG. **5A**) and extended (FIG. **5B**) positions respectively. As can be seen in FIG. **5A**, even in the retracted position, tubular member **300** may still be slightly stretched and exert a biasing force to press the needle shaft portion **220** so that the needle tip portion **210** of the needle bundle **200** is pushed towards or against a side wall of the housing **40**.

(74) As shown in FIG. **5B**, the middle tubular section **330** of the tubular member **300** is further stretched and tensioned at the extended position by a length X , and thus produces a relatively large biasing force. Conveniently, the helical ridge **340** when stretched produces a biasing force $\{\text{right arrow over (F)}\}$, generally directed along the helical direction of the helical ridge **340**, as illustrated in FIG. **5B**. The biasing force $\{\text{right arrow over (F)}\}$ has a longitudinal component $\{\text{right arrow over (F)}\}.\text{sub.1}$ in the axial direction A of channel **120** and the needle shaft portion **220**, and a lateral component $\{\text{right arrow over (F)}\}.\text{sub.2}$ in a lateral direction generally perpendicular to the

axial direction A, where {right arrow over (F)}={right arrow over (F)}.sub.1+{right arrow over (F)}.sub.2, as illustrated in FIG. 6. The longitudinal component {right arrow over (F)}.sub.1 pulls the needle bundle **200** upward towards the retracted position, and the lateral component {right arrow over (F)}.sub.2 presses the tip end of the needle bundle **200** towards the guiding wall **110**.

(75) For a typical tattoo machine, the distance X between the retracted and extended positions of needle bundle **200** may be about 2 mm to about 5 mm, as the driving shaft of the needle actuator in a typical tattoo machine can move the same distance during each stroke.

(76) As can be appreciated, in the retracted position as shown in FIG. 5A, the tubular member **300** may still be tensioned. For example, when the needle module **10** is not connected to the handle of a tattoo machine (not shown), the tubular member **300** may be tension free and in a relaxed state. However, when housing **40** of the needle module **10** is coupled to or connected to the handle of the tattoo machine, the needle shaft portion **220** may be initially pushed downward such as by about 2 mm to about 4 mm relative to the housing body portion **100**. This downward movement of the needle shaft portion **220** tensions the tubular member **300** even though the needle bundle **200** is still in the retracted position initially. During use, the needle shaft portion **220** and needle bundle **200** may be pushed to move further downward by about 2 mm to about 5 mm.

(77) As discussed earlier, the annular rim **420** and groove **430** of cap **400** engages the upper end portion **320** of tubular member **300**, and upper end portion **320** is secured in position between the cap **400** and the inner wall at the upper end of channel **120**, so that upper end portion **320** cannot move relative to the housing body portion **100** in the longitudinal direction (along axis A) during operation while the needle bundle **200** reciprocally moves up and down relative to the body portion **100**. The upper end portion **320** of tubular member **300** is also configured to abut and resiliently grip the inner wall of the housing body portion **100** and to provide a tight seal between the upper end portion **320** and the housing body portion **100**.

(78) As better illustrated in FIGS. 7B, 7C, 8A, and 8B, two symmetrically arranged helical ridges **340** may be provided. Each ridge may be rib-shaped.

(79) The lower opening **150** in housing **40** may have different shapes and sizes. For example, if the needle bundle **200** in the needle module **10** is a flat needle bundle with a row of needles as illustrated in FIGS. 2, 3 and 4, the opening **150** may have a corresponding rectangular profile. Further, the needle guiding wall **110** may have a corresponding flat inner surface, also as illustrated in FIGS. 3 and 4. In the embodiment as depicted in FIG. 7B, the symmetry plane B passes through and is parallel to the central axis A and is perpendicular to the guiding wall **110** of the housing body portion **100**. Two helical ridges **340** are arranged symmetrically with respect to the symmetry plane B.

(80) In different embodiments, the needle tip portion **210** of the needle bundle **200** may have a generally cylindrical overall profile, in which case the opening **150** of the housing **40** may be circular. The inner wall of the guiding wall **110** may also be curved. The needle bundle **200** may contact the guiding wall **110** along a longitudinal line, which is referred to the needle guiding line herein. In this case, the symmetrical plane B may contain both the axis A and the needle guiding line.

(81) As noted earlier, in different embodiments, the helical ridges **340** on tubular member **300** may have different shapes and configurations.

(82) For example, in some embodiments, as shown in FIG. 9A, a helical ridge **340** may have curved side edges when viewed in the side elevation view. In other words, the edges of the helical ridge **340** in some embodiments may be curved in the face-on elevation view, when the viewer is directly facing the helical ridge **340**.

(83) In some embodiments, a helical ridge **340** may have generally parallel side edges as shown in FIG. 7A or may have unparallel side edges as shown in FIG. 9A.

(84) In some embodiments, as shown in FIG. 9B, a helical ridge **340** may have helically aligned step-shaped side edges. That is, the edges of the helical ridge **340** in some embodiments may be

step-shaped or zig-zap shaped in the face-on elevation view. In alternative embodiments, such as when the tubular member **340** has a flat side surface, the step-shaped side edges may also be aligned along a straight or curved line that extends on the flat surface in both the longitudinal direction and the lateral perimetral direction.

(85) Two or more helical ridges **340** may also be arranged side-by-side, and generally parallel to one another, as illustrated in FIG. **9C**. The two or more helical ridges **340** may be positioned on one side of the tubular member **300**.

(86) As illustrated in FIGS. **7A** and **9D**, the helical ridges **340** may have different widths in different embodiments. The helical ridges **340** may also have different heights in different embodiments. As can be appreciated, a wider or taller ridge may provide a larger biasing force.

(87) FIGS. **9D**, **9E** and **9F** illustrate that a helical ridge **340** may have different widths in different embodiments, and a helical ridge **340** may be positioned at different positions. For example, as shown in FIG. **9D**, the helical ridge **340** has a relatively large width and the thinner wall area **332** is much larger in area size than the thinner wall area **334**. In FIG. **9E**, the middle tubular section **330** of tubular member **300** has only different regions, the thinner wall area **332** and the thicker helical ridge **340** (the area **334** does not exist or has a size of zero), which are separated by the helical boundary **345**. In FIG. **9F**, the thinner wall area **332** has a larger area size as compared to FIG. **9E**, and the helical ridge **340** has a correspondingly smaller area size as compared to FIG. **9E**. FIGS. **10A** and **10B** show perspective views of tubular member **300** shown in FIGS. **9E** and **9F** respectively.

(88) In the embodiment of the tubular member **300** shown in FIG. **9G**, the wall **332** and wall **334** have different wall thicknesses. For example, wall **332** may be thinner than wall **334**. In this case, when tubular member **300** is stretched, walls **332** and **334** produce different biasing forces, the force produced by wall **334** being larger than that produced by wall **332**. Such force differential affects the total lateral biasing force, in addition to the effects produced by the helical ridge **340**.

(89) FIG. **9H** illustrates an embodiment of the tubular member **300** where the tubular wall **332** in the middle section is corrugated or is a bellows section. Wall **334** in FIG. **9H** is not corrugated. FIG. **9I** illustrates an embodiment of the tubular member **300** where both the wall **332** and wall **334** are corrugated or bellows sections.

(90) As can be appreciated, when one or more areas of the middle tubular section **330** has curved or corrugated walls, the walls at different areas may have different rectified lengths, which are longer than a straight wall. For a curved line, its rectified length refers to the length of the curve that has been rectified. When rectified, the curve gives a straight line segment with the same length as the curve's arc length. For a straight line, its rectified length is the same as the length of the straight line.

(91) In the embodiment shown in FIG. **9H**, wall **334** has a shorter rectified length than wall **332** so that wall **334** would be tensioned more than wall **332** when the middle tubular section **330** is stretched due to downward movement of the needle bundle **200**.

(92) As can be appreciated, in the embodiment of FIG. **9H**, the corrugated wall **332** is less tensioned than the non-corrugated wall **334** when the tubular member **300** is stretched. Thus, the longitudinal forces produced by walls **332** and **334** are not balanced and the imbalance produces an additional lateral biasing force.

(93) A helical ridge **340** may have any number of cross-sectional shapes or profiles, some of which are illustrated in FIGS. **11A** to **11F**. For example, the cross-section profile of a helical ridge **340** may be generally polygonal or semicircular. The polygonal cross-section may be square, as illustrated in FIG. **11A**; trapezoidal, as illustrated in FIG. **11B**; triangular, as illustrated in FIG. **11D**; or rectangular, as illustrated in FIGS. **11E** and **11F**. The cross-section profile of a helical ridge **340** may also include a semicircular edge as illustrated in FIG. **11C**.

(94) In FIGS. **11A-11E**, wall **332** and wall **334** have the same wall thickness. In FIG. **11F**, wall **334** is thicker than wall **332**. In any event, the wall thickness at helical ridge **340** is larger than at both

walls **332** and **334**.

(95) In FIGS. **11E** and **11F**, the helical ridge **340** has a generally rectangular cross-sectional profile, but the top edge may be curved or arched. For example, the outer top edge of helical ridge **340** may be concentric with the tubular section formed by walls **332** and **334**.

(96) Conveniently, the helical ridge **340** or different portions having a helical boundary and different thickness, when stretched, can provide a biasing force that biases the needle bundle both longitudinally towards the retracted position and laterally towards a lateral side of the channel **120**.

(97) Tubular member **300** also conveniently provides a fluid seal between the upper open end and lower open end of the housing channel **120**.

(98) As compared with the biasing member disclosed in U.S. Ser. No. 11/052,232 or the elastic membrane disclosed in CN 106902452A, a tubular member as described herein provides a number of beneficial effects or advantages.

(99) For example, a tubular member with one or more helical ridges as described herein is easier to manufacture, as compared to an elastic tubing with opposite sides of different rectified lengths and corrugated tubular sections. For instance, when manufactured by molding, the internal shapes of the molds for producing the tubular member **300** illustrated in FIGS. **7A**, **7C**, and **7D** are not as complex as in a mold for producing a biasing member with different rectified lengths and corrugated tubular sections. As a result, the production efficiency and product yield can be improved. Further, it is easier to inspect small defects on the external surface of a product with simpler external surface features.

(100) The elastic membrane disclosed in CN 106902452A have regions of different thicknesses and the boundaries between the different regions extend in the axial or longitudinal direction. The elastic membrane of CN 106902452A may have an elastic rib extending along the direction of the needle movement on one side of the elastic membrane. A drawback of the elastic membrane disclosed in CN 106902452A is that it is difficult to adjust amounts, or ratio, of the generated forces in different directions. For example, while the thicknesses of the different portions of the membrane may be adjusted to generate different force ratios, a large thickness difference in the same membrane can result in manufacturing complexity and reduction in performance and product durability.

(101) As compared to a region boundary or elastic rib that extends linearly along the axial or longitudinal direction of the needle, a helical ridge or boundary as described herein can provide additional lateral forces that can be more easily adjusted by changing the helical direction without changing the thickness of the helical ridge or any portion of the tubular membrane. For example, a higher lateral to longitudinal biasing force ratio may be achieved with a helical ridge or boundary without changing the thicknesses of the tubular member.

(102) The needle module **10** may be a single-use disposable module, and can be used directly after opening the needle module packaging without further cleaning, sanitization, or sterilization, and can be disposed after a single use without cleaning or any other treatment.

(103) For clarity, it is noted that “single use” may refer to use of a needle or needle module for one complete operation on a single individual subject. During this operation, different needle modules may be used to apply different ink colors or for different purposes. For example, it may be typical to use two to five different types of needles during a single operation on a subject, depending on the complexity of the design to be applied.

(104) As can be appreciated, the tubular member **300** is configured to provide both a longitudinal biasing force and a lateral or radial biasing force, with the provision of a helical ridge or helical boundary.

(105) It is noted that a tubular member with a pair of corresponding helical or slanted ridges arranged symmetrically with respect to the sagittal plane provides certain benefits over a single ridge extending along the sagittal plane. For example, as compared to single ridge in the sagittal plane, the pair of symmetrical ridges can be made thicker or larger and provide a stronger radial

biasing force without interfering with the guiding wall **110**. This is particularly helpful when the space in the channel **120** is limited so the tubular member **300** needs to be small in size with relatively thin walls.

(106) As now can be appreciated, during operation, the tubular member **300** also separates the lower open end of body portion **100** from the upper open end of body portion **100**, and provides a fluid-tight seal in the channel **120** between the lower end and the upper end of the housing **40** such that bodily fluids exiting from the punctured skin of the subject being treated will be prevented from travelling from the needle tip portion **210** to the upper end of the housing through the inner channel **120**. The seal also prevents ink from entering and passing through the upper end of needle housing **40**. The seal thus conveniently prevents the subject's bodily fluids and ink from contacting parts of the needle handle (not shown) or tattoo machine (not shown) connected to the needle module **10**. Conveniently, the tattoo machine and the needle handle may be reused after each treatment or after changing the needle module, without the need to re-sterilize any part of the tattoo machine. In some cases, the needle handle may also be disposable.

(107) It can now also be appreciated that the above convenient effects can also be achieved with different embodiments or variations of tubular member **300** as depicted in the drawings. For example, as alluded to earlier, the tubular member **300** may have a polygonal cross-section and may include slanted ridges or boundaries on its external surface, which replace the helical boundaries or ridges, but still provide similar biasing effects as discussed herein.

(108) As can be appreciated, a needle module or needle assembly described herein may be used or adapted to apply other types of liquids to skin. For example, the applied liquid may include colored liquids or pigments, or may include a medicinal or therapeutic agent, collagen, or other like or similar substances. The needle assembly may be used in a liquid applicator for applying the selected liquid.

(109) Other features, modifications, and applications of the embodiments described here may be understood by those skilled in the art in view of the disclosure herein.

Concluding Remarks

(110) It will be understood that any range of values herein is intended to specifically include any intermediate value or sub-range within the given range, and all such intermediate values and sub-ranges are individually and specifically disclosed.

(111) It will also be understood that the word “a” or “an” is intended to mean “one or more” or “at least one”, and any singular form is intended to include plurals herein.

(112) It will be further understood that the term “comprise”, including any variation thereof, is intended to be open-ended and means “include, but not limited to,” unless otherwise specifically indicated to the contrary.

(113) When a list of items is given herein with an “or” before the last item, any one of the listed items or any suitable combination of two or more of the listed items may be selected and used.

(114) Of course, the above described embodiments of the present disclosure are intended to be illustrative only and in no way limiting. The described embodiments are susceptible to many modifications of form, arrangement of parts, details, and order of operation. The invention, rather, is intended to encompass all such modification within its scope, as defined by the claims.

LIST OF REFERENCES AND CORRESPONDING ELEMENTS AS SHOWN IN THE FIGURES

(115) TABLE-US-00001 Reference Corresponding element 10 Needle module 40 Housing 100 Body portion 110 Guiding wall 120 Channel 122 Upper open end 124 Lower open end 130 Coupling rib 150 (lower end) Opening 200 Needle Bundle 210 Needle Tip portion 215 Connection rod 220 Needle shaft portion 222 Bore 230 Shoulder 232 Collar 234 Neck section 300 Tubular member 310 Lower end portion 312 Central opening 320 Upper end portion 322 Annular ridges 330 Middle tubular section 332 First wall 334 Second wall 340 Helical ridge 345 Boundary 400 Cap 410 Tube 420 Annular rim 430 (lower) Groove 440 (upper) Groove 450 (central) Opening A-A Central (longitudinal) axis B-B Symmetry (sagittal) plane 7D-7D Frontal plane

Claims

1. A needle assembly for a liquid applicator, comprising: a housing comprising a longitudinal channel, the channel comprising an upper open end and a lower open end; a needle bundle movably mounted in the channel, the needle bundle configured and mounted to be driven by a driving shaft to reciprocally move along a longitudinal direction between a retracted position and an extended position; an elastic tubular member comprising a first end attached to the housing and a second end attached to the needle bundle to bias the needle bundle to move longitudinally towards the retracted position, the tubular member comprising a first portion and a second portion thinner than the first portion, wherein a slanted boundary between the first portion and the second portion extends between the first end and the second end in both the longitudinal direction and a lateral perimetral direction of the tubular member to provide a lateral biasing force that biases the needle bundle laterally toward a lateral side of the channel.
2. The needle assembly of claim 1, wherein the slanted boundary comprises a pair of slanted boundaries, arranged symmetrically with respect to a plane passing through a longitudinal axis of the tubular member.
3. The needle assembly of claim 1, wherein the first portion of the tubular member comprises a slanted ridge extending in both the longitudinal direction and the lateral perimetral direction for providing the lateral biasing force.
4. The needle assembly of claim 3, wherein the slanted ridge extends from the first end to the second end of the tubular member.
5. The needle assembly of claim 3, wherein the slanted ridge comprises a pair of slanted ridges arranged symmetrically with respect to an axial plane passing through a longitudinal axis of the tubular member.
6. The needle assembly of claim 5, wherein the pair of slanted ridges comprises a pair of helical ridges.
7. The needle assembly of claim 3, wherein the slanted ridge has a generally polygonal or semicircular cross-section.
8. The needle assembly of claim 7, wherein the polygonal cross-section is triangular, rectangular, or trapezoidal.
9. The needle assembly of claim 1, wherein an external lateral perimeter of the tubular member is generally polygonal.
10. The needle assembly of claim 1, wherein the polygonal lateral perimeter has 3 to 12 generally straight sections.
11. The needle assembly of claim 1, wherein an external lateral perimeter of the tubular member is generally elliptical.
12. The needle assembly of claim 1, wherein the tubular member has a generally cylindrical, frustum, or prismatic profile.
13. The needle assembly of claim 1, wherein the slanted boundary comprises a straight boundary.
14. The needle assembly of claim 1, wherein the slanted boundary comprises a curved boundary.
15. The needle assembly of claim 1, wherein the first portion and the second portion of the tubular member are separated by a step-shaped side wall at the slanted boundary.
16. The needle assembly of claim 15, wherein the step-shaped side wall comprises parallel side edges arranged along the slanted boundary.
17. The needle assembly of claim 1, wherein the first portion and the second portion of the tubular member are separated by a curved side wall at the slanted boundary.
18. The needle assembly of claim 1, wherein the second portion of the tubular member comprises a bellows section.
19. The needle assembly of claim 1, wherein the first and second ends of the tubular member are

sealingly attached to the housing and the needle bundle to prevent fluid communication between the lower open end and the upper open end of the housing.

20. The needle assembly of claim 1, wherein the tubular member comprises silicone, latex, or rubber.
